MA3236 NONLINEAR PROGRAMMING
Semester 1, AY 2026/2027
Tutorial 9

Review: Lagrangian duality (L18)

Dual problem. All problem functions are finite on $\mathbf{R}^n$, and all constraints are explicit:

$$p^* = \inf\{f(\mathbf{x}) : \mathbf{g}(\mathbf{x}) = \mathbf{0},\ \mathbf{h}(\mathbf{x}) \le \mathbf{0}\}, \qquad W = \mathbf{R}^m \times \mathbf{R}^p_+.$$

The equality multipliers are free; inequality multipliers are nonnegative. Set

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x} \in \mathbf{R}^n} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^T \mathbf{h}(\mathbf{x}), \qquad d^* = \sup_{(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in W} \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}).$$

For fixed multipliers, take the *global* inner infimum over $\mathbf{R}^n$.

Weak duality and objective certificate. For primal feasible $\mathbf{x}$ and $\mathbf{w} = (\boldsymbol{\lambda}, \boldsymbol{\nu}) \in W$,

$$\theta(\mathbf{w}) \le L(\mathbf{x}, \mathbf{w}) \le f(\mathbf{x}) \quad \Longrightarrow \quad \theta(\mathbf{w}) \le d^* \le p^* \le f(\mathbf{x}).$$

- No convexity or constraint qualification is needed. When the values are finite, distinguish the optimal gap $p^* - d^*$ from the computable gap $G = f(\mathbf{x}) - \theta(\mathbf{w})$:

  $$0 \le \max\{p^* - d^*,\ f(\mathbf{x}) - p^*,\ d^* - \theta(\mathbf{w})\} \le G.$$

  Thus $G = 0$ certifies both global optima; $G \le \varepsilon_{\text{out}}$ certifies objective accuracy.

- If $p^* = -\infty$, then $\theta = -\infty$ throughout W. If $d^* = +\infty$, the primal problem is infeasible. The converses need not hold.

Effective domain, finiteness, attainment, uniqueness.

$$\operatorname{dom} \theta = \{\mathbf{w} : \theta(\mathbf{w}) > -\infty\}, \qquad \mathcal{F}_D = W \cap \operatorname{dom} \theta.$$

The pointwise infimum of affine functions is concave, so θ is concave even for a nonconvex primal problem. When the inner problem is separable, minimize its terms individually. In particular,

$$\inf_{z \in \mathbf{R}} (az^2 + bz + c) = \begin{cases} c - b^2/(4a), & a > 0, \\ c, & a = b = 0, \\ -\infty, & a < 0 \text{ or } (a = 0,\ b \neq 0). \end{cases}$$

A finite infimum may be unattained or have many minimizers. Likewise, the dual supremum may be unattained or have many maximizers. The dual function may be nonsmooth.

Review: strong duality and saddle points (L19)

Strong duality. Strong duality means $d^* = p^* \in \mathbf{R}$. Since the multiplier supremum enforces the constraints, this is an exchange of optimization order:

$$p^* = \inf_{\mathbf{x}\in\mathbf{R}^n} \sup_{\mathbf{w}\in W} L(\mathbf{x},\mathbf{w}) \;=\; \sup_{\mathbf{w}\in W} \inf_{\mathbf{x}\in\mathbf{R}^n} L(\mathbf{x},\mathbf{w}) = d^*.$$

For convex f, h_j and affine $\mathbf{g}(\mathbf{x}) = \mathbf{B}\mathbf{x}-\mathbf{b}$, with $p^* \in \mathbf{R}$, the convex strong-duality theorem gives

$$\underbrace{\exists\, \widehat{\mathbf{x}}:\ \mathbf{B}\widehat{\mathbf{x}} = \mathbf{b},\quad h_j(\widehat{\mathbf{x}}) < 0\ (\forall j)}_{\text{Slater's condition}} \quad\Longrightarrow\quad d^* = p^* \text{ and a dual optimizer exists.}$$

- Convexity and Slater are sufficient, not necessary. Convexity or Slater alone does not ensure zero gap.
- Primal attainment is a separate issue. For example, $\inf_{x\geq 0} e^{-x} = 0$ has no minimizer, although convexity and Slater hold.
- For linear programming, strict feasibility is unnecessary:

  $$\min\{\mathbf{c}^T\mathbf{x} : \mathbf{B}\mathbf{x} = \mathbf{b},\ \mathbf{x}\geq\mathbf{0}\} \quad\leftrightarrow\quad \max\{-\mathbf{b}^T\boldsymbol{\lambda} : \mathbf{c}+\mathbf{B}^T\boldsymbol{\lambda}\geq\mathbf{0}\}.$$

  If either side is feasible with finite optimal value, both optima are attained and equal. An unbounded side forces the other side to be infeasible.

Global saddle point. A global saddle point $(\mathbf{x}^*, \mathbf{w}^*)$, with $\mathbf{w}^* \in W$, satisfies

$$L(\mathbf{x}^*,\mathbf{w}) \leq L(\mathbf{x}^*,\mathbf{w}^*) \leq L(\mathbf{x},\mathbf{w}^*) \qquad \forall\ \mathbf{x}\in\mathbf{R}^n,\quad \mathbf{w}\in W.$$

These inequalities are equivalent to primal and dual optimality of the pair with $f(\mathbf{x}^*) = \theta(\mathbf{w}^*)$. This equivalence needs no convexity. For differentiable data, write KKT as

$$\nabla_{\mathbf{x}} L = \mathbf{0},\quad \mathbf{g} = \mathbf{0},\quad \mathbf{h}\leq\mathbf{0},\quad \boldsymbol{\nu}\geq\mathbf{0},\quad \nu_j h_j = 0\ (\forall j).$$

- Attained primal and dual optima with zero gap imply KKT. Conversely, KKT implies a global saddle point when f, h_j are convex and $\mathbf{g}$ is affine, because stationarity then globally minimizes $L(\cdot, \mathbf{w}^*)$.
- In a nonconvex problem, KKT alone does not certify zero gap. Even a zero gap alone supplies no saddle point unless both optima are attained.

Primal recovery. Even at a dual optimizer, an arbitrary inner minimizer may be infeasible. Choose $\mathbf{x}^* \in \arg\min_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ and verify

$$\mathbf{g}(\mathbf{x}^*) = \mathbf{0},\quad \mathbf{h}(\mathbf{x}^*)\leq\mathbf{0},\quad (\boldsymbol{\nu}^*)^T\mathbf{h}(\mathbf{x}^*) = 0 \quad\Longrightarrow\quad f(\mathbf{x}^*) = \theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*).$$

1. **A quadratic dual with two kinds of multipliers.** Consider

$$\begin{aligned} \min_{\mathbf{x}\in\mathbf{R}^2} \quad & f(\mathbf{x}) = 2x_1^2 + x_2^2 - 4x_1 - 6x_2 \\ \text{s.t.} \quad & g(\mathbf{x}) = x_1 + x_2 - 4 = 0, \\ & h(\mathbf{x}) = -x_1 + 2x_2 - 4 \le 0. \end{aligned}$$

   (a) Write $L(\mathbf{x}, \lambda, \nu)$, state the dual multiplier restrictions, and derive $\theta(\lambda, \nu)$.

   (b) Solve the dual problem and recover a primal minimizer. Check if strong duality holds by exhibiting a Slater point.

   (c) Verify the saddle inequalities directly at this primal–dual pair.

Solution. The equality multiplier λ is free and the inequality multiplier satisfies $\nu \ge 0$. The Lagrangian is

$$L = 2x_1^2 + (\lambda - \nu - 4)x_1 + x_2^2 + (\lambda + 2\nu - 6)x_2 - 4\lambda - 4\nu.$$

Its Hessian in $\mathbf{x}$ is $\operatorname{diag}(4, 2) \succ 0$, so its unique minimizer is

$$x_1(\lambda, \nu) = 1 - \frac{\lambda}{4} + \frac{\nu}{4}, \qquad x_2(\lambda, \nu) = 3 - \frac{\lambda}{2} - \nu.$$

Substitution gives the finite dual function on all of $\mathbf{R}^2$:

$$\theta(\lambda, \nu) = -\frac{3}{8}\lambda^2 - \frac{3}{4}\lambda\nu - \frac{9}{8}\nu^2 + \nu - 11.$$

To maximize it subject to $\nu \ge 0$, solve

$$-\frac{3}{4}\lambda - \frac{3}{4}\nu = 0, \qquad -\frac{3}{4}\lambda - \frac{9}{4}\nu + 1 = 0.$$

The solution is $\lambda^* = -2/3$, $\nu^* = 2/3$. The dual Hessian is negative definite, so this feasible stationary point is the unique dual maximizer. Recovery gives

$$\mathbf{x}^* = (4/3, 8/3)^T, \qquad f(\mathbf{x}^*) = \theta(\lambda^*, \nu^*) = -32/3.$$

Both displayed constraints are active at $\mathbf{x}^*$, and weak duality certifies both optima. The point $(2, 2)^T$ satisfies $g = 0$ and $h = -2 < 0$, so Slater also holds. Together with the convexity of f and h, strong duality holds.

For any $\lambda \in \mathbf{R}$ and $\nu \ge 0$, feasibility and activity give

$$L(\mathbf{x}^*, \lambda, \nu) = f(\mathbf{x}^*) = L(\mathbf{x}^*, \lambda^*, \nu^*).$$

For any $\mathbf{x} \in \mathbf{R}^2$, completing the square gives

$$L(\mathbf{x}, \lambda^*, \nu^*) = L(\mathbf{x}^*, \lambda^*, \nu^*) + 2(x_1 - 4/3)^2 + (x_2 - 8/3)^2.$$

The nonnegative square terms prove the other saddle inequality.

2. **Constraint formulation, KKT, and the duality gap.** Let $f(x) = x^4 - 2x^2$ for $x \in \mathbf{R}$.

   (a) Consider $\min_x f(x)$ subject to $x^2 \leq 0$. Find p^*, derive $\theta(\nu)$ for every $\nu \geq 0$, and determine d^* and all optimal multipliers. Check whether the convex strong-duality theorem applies.

   (b) Replace the inequality by $x = 0$. Find the dual optimal value and all optimal multipliers, and compare the two duality gaps. Determine whether $(x, \lambda) = (0, 0)$ satisfies KKT and whether a global Lagrangian saddle point exists for this equality formulation.

Solution. (a) The only feasible point is $x = 0$, so $p^* = 0$. For $\nu \geq 0$,

$$L(x, \nu) = x^4 + (\nu - 2)x^2.$$

Writing $z = x^2 \geq 0$ reduces the inner problem to $\min_{z \geq 0}\{z^2 + (\nu - 2)z\}$. Its minimizer is $z = (2 - \nu)/2$ when $0 \leq \nu < 2$, and $z = 0$ when $\nu \geq 2$. Consequently,

$$\theta(\nu) = \begin{cases} -(2 - \nu)^2/4, & 0 \leq \nu < 2, \\ 0, & \nu \geq 2. \end{cases}$$

Hence $d^* = 0 = p^*$ and the optimal multipliers are exactly $[2, \infty)$. The objective is nonconvex ($f''(0) = -4$), and $x^2 < 0$ is impossible, so both convexity and Slater fail. Matching values nevertheless certify strong duality. In particular, $(x, \nu) = (0, 2)$ is a global saddle point: $L(0, \nu) = 0$ for every $\nu \geq 0$, while $L(x, 2) = x^4 \geq 0$.

(b) The feasible set is again $\{0\}$, so $p^* = 0$. Now $L(x, \lambda) = x^4 - 2x^2 + \lambda x$, with $\lambda \in \mathbf{R}$. Testing $x = 1$ and $x = -1$ bounds its infimum:

$$\theta(\lambda) \leq \min\{-1 + \lambda, -1 - \lambda\} = -1 - |\lambda| \leq -1.$$

At $\lambda = 0$, $x^4 - 2x^2 = (x^2 - 1)^2 - 1 \geq -1$, with equality at $x = \pm 1$. Thus $\theta(0) = -1$, and $\lambda^* = 0$ is the unique dual maximizer. The gap is $p^* - d^* = 1$, compared with zero in (a).

The KKT equations are $4x^3 - 4x + \lambda = 0$ and $x = 0$, so $(0, 0)$ is a KKT pair. It is not a global saddle point: $L(1, 0) = -1 < L(0, 0) = 0$. The positive gap rules out every global saddle point. Thus identical primal feasible sets can give different Lagrangian dual bounds when their constraint formulations differ.

3. **A dual with a restricted effective domain.** Consider

$$\min_{\mathbf{x}\in\mathbf{R}^3} f(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 \quad \text{s.t.} \quad g(\mathbf{x}) = 2x_1 + x_2^2 + x_3 - 4 = 0.$$

(a) Derive $\theta(\lambda)$, including every λ for which $\theta(\lambda) = -\infty$.

(b) Solve the dual problem and determine its Lagrangian-subproblem minimizer set at λ^*.

(c) Recover all primal global minimizers and certify zero gap. Explain why the usual convex Slater theorem is not the justification.

Solution. The equality multiplier is free, and

$$L(\mathbf{x}, \lambda) = x_1^2 + (1+\lambda)x_2^2 + x_3^2 + 2\lambda x_1 + \lambda x_3 - 4\lambda.$$

If $\lambda < -1$, the negative x_2^2 coefficient makes the infimum $-\infty$. Otherwise, minimizing in x_1, x_3 gives $x_1 = -\lambda$, $x_3 = -\lambda/2$. When $\lambda > -1$, $x_2 = 0$; when $\lambda = -1$, every x_2 minimizes its term. Thus

$$\theta(\lambda) = \begin{cases} -\dfrac{5}{4}\lambda^2 - 4\lambda, & \lambda \geq -1, \\ -\infty, & \lambda < -1. \end{cases}$$

The finite branch has derivative $-(5/2)\lambda - 4 < 0$ on $[-1, \infty)$. Therefore $\lambda^* = -1$ and $d^* = 11/4$.

At λ^* the minimizing set is $\{(1, t, 1/2)^T : t \in \mathbf{R}\}$. Feasibility requires $t^2 = 3/2$. The two recovered points are

$$\mathbf{x}_\pm^* = (1, \pm\sqrt{3/2}, 1/2)^T, \qquad f(\mathbf{x}_\pm^*) = 11/4 = \theta(-1).$$

Weak duality proves their global optimality and zero gap. Conversely, any primal minimizer must minimize $L(\cdot, -1)$, since $L = f$ on the feasible set. Hence these are all minimizers. The equality is nonlinear, so the convex-program theorem requiring affine equalities does not apply; the matching values are the certificate.